

Advanced Dynare

Chapter VIII

Higher-Order Perturbation

Johannes Pfeifer

University of the Bundeswehr Munich

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Outline

1. The Perturbation Approach

The Idea

Regular vs. singular perturbation

2. Higher-order perturbation

Introduction

First-order approximation

Second-order solution

Terminology: steady states and ergodic distribution

3. IRFs

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The idea of regular perturbation

- Consider problem that reduces to solving

$$f(x, \varepsilon) = 0 \quad (1)$$

for variable x and a parameter ε

- Assume each value for ε , equation has solution for x
→ set of equations in x parameterized by ε

- Let $x(\varepsilon)$ be smooth function of ε :

$$f(x(\varepsilon), \varepsilon) = 0 \quad (2)$$

- Goal: find $x(\varepsilon)$
- Generally impossible to solve (2) for arbitrary ε , but may be possible for special ε

Simple example

- Assume $x(0)$ is known and use implicit differentiation in univariate/scalar case:

$$f_x(x(\varepsilon), \varepsilon) x_\varepsilon(\varepsilon) + f_\varepsilon(x(\varepsilon), \varepsilon) = 0 \quad (3)$$

- Equation implicitly defines derivative $x_\varepsilon(\varepsilon)$
- Problem: still depends on unknown $x(\varepsilon)$
- Evaluating at $\varepsilon = 0$ yields

$$f_x(x(0), 0) x_\varepsilon(0) + f_\varepsilon(x(0), 0) = 0 \quad (4)$$

which implies

$$x_\varepsilon(0) = -\frac{f_\varepsilon(x(0), 0)}{f_x(x(0), 0)} \quad (5)$$

- Given we know $x(0)$ and function f , we can compute $x_\varepsilon(0)$ and get the linear approximation

$$x(\varepsilon) = x(0) - \frac{f_\varepsilon(x(0), 0)}{f_x(x(0), 0)}(\varepsilon - 0) \quad (6)$$

Continuing the simple example

- Differentiating a second time yields

$$\begin{aligned} 0 = & f_{xx}(x(\varepsilon), \varepsilon) (x_\varepsilon(\varepsilon))^2 + 2f_{x\varepsilon}(x(\varepsilon), \varepsilon) x_\varepsilon(\varepsilon) \\ & + f_x(x(\varepsilon), \varepsilon) x_{\varepsilon\varepsilon}(\varepsilon) + f_{\varepsilon\varepsilon}(x(\varepsilon), \varepsilon) \end{aligned} \quad (7)$$

- This implies

$$\begin{aligned} x_{\varepsilon\varepsilon}(0) = & (-1) \times \\ & \frac{f_{xx}(x(0), 0) x_\varepsilon^2(0) + 2f_{x\varepsilon}(x(0), 0) x_\varepsilon(0) + f_{\varepsilon\varepsilon}(x(0), 0)}{f_x(x(0), 0)} \end{aligned}$$

- Evaluating at 0 yields:

$$x(\varepsilon) = x(0) - x_\varepsilon(0)(\varepsilon - 0) - \frac{1}{2}x_{\varepsilon\varepsilon}(0)(\varepsilon - 0)^2 \quad (8)$$

- As long as there are sufficient derivatives of f , we can continue
- Each successive differentiation results in a linear equation for the next derivative of $x(\varepsilon)$
- Effectively, **we construct our solution from a point that we know a lot about**, usually the deterministic steady state

Singular perturbation

- The example up to this point considered **regular perturbation**, but we are typically interested in stochastic problems that depend on shocks ν_t :

$$E_t f(x_t, \nu_t) = 0 \quad (9)$$

- But we only know the solution at the deterministic steady state, i.e. in the deterministic case with

$$f(x_t, 0) = 0 \quad (10)$$

- We could try to use our previous approach by writing:

$$E_t f(x_t, \varepsilon \nu_t) = 0 \quad (11)$$

and perturbing around $\varepsilon = 0$

- Problem: changing ε from 0 to any non-zero value introduces a **singular perturbation** as it changes the order of the problem
- E.g. $x_t = \rho x_{t-1} + \varepsilon \nu_t$ would collapse to $x_t = x_{t-1} = \bar{x}$

Singular perturbation

- Fleming (1971): we can treat this singular perturbation in ε as a regular perturbation (according to Judd 1998)
- However: not completely clear, what the correct way is
- It looks as if Fleming (1971) first takes the derivative w.r.t. to the perturbation parameter and then tries to find the solution to the perturbed system
- Standard practice in economics is to simultaneously take the derivative w.r.t. to the perturbation parameter and the state variables
- For example, Fernández-Villaverde and Rubio-Ramírez (2010) add perturbation parameter to the state vector and treat it essentially as a state
- This creates a lot of issues

Example

- Consider the problem

$$\max_{y_t} -E_t \left(y_t - e^{x_{t+1}} \right)^2 \quad (12)$$

- Can be motivated by the optimization problem of a bank choosing its capital level y_t for next period, while the capital requirement $e^{x_{t+1}}$ is stochastic
- Having more or less capital than this target level is costly
- Assume that the required capital follows:

$$x_{t+1} = \rho x_t + \varepsilon_{t+1} \quad (13)$$

- The FOC is given by

$$y_t = E_t \left[e^{\rho x_t + \varepsilon_{t+1}} \right] \quad (14)$$

and the analytical solution by:

$$y_t = e^{\rho x_t + \frac{1}{2} \text{Var}(\varepsilon_t)} \quad (15)$$

Singular and regular perturbation simultaneously

- Take the FOC

$$y_t = E_t \left[e^{\rho x_t + \sigma \varepsilon_{t+1}} \right] \quad (14)$$

and perform a Taylor approximation w.r.t. x_t and σ about the deterministic steady state $(0, 0)$

$$\begin{aligned} y_t &\approx E_t \left[1 + e^0 \rho (x_t - 0) + e^0 \varepsilon_{t+1} (\sigma - 0) \right] \\ &= 1 + \rho x_t \end{aligned} \quad (16)$$

- Good approximation to the true solution if system is deterministic, i.e. $Var(\varepsilon_t) = 0$
- Solution is certainty equivalent (as opposed to the true solution/problem)

Singular perturbation first

- Example follows the procedure advocated by Evers (2012)
- Take the FOC

$$y_t = E_t \left[e^{\rho x_t + \sigma \varepsilon_{t+1}} \right] \quad (14)$$

and perform a second order Taylor approximation w.r.t. σ about the deterministic steady model ($\sigma = 0$):

$$\begin{aligned} y_t &\approx E_t \left[e^{\rho x_t} + e^{\rho x_t} \varepsilon_{t+1} (\sigma - 0) + e^{\rho x_t} \varepsilon_{t+1}^2 (\sigma - 0)^2 \right] \\ &= e^{\rho x_t} + \frac{1}{2} e^{\rho x_t} \sigma^2 E_t \varepsilon_{t+1}^2 \\ &= e^{\rho x_t} + \frac{1}{2} e^{\rho x_t} \text{Var}_t (\varepsilon_{t+1}) = e^{\rho x_t} + \frac{1}{2} e^{\rho x_t} \text{Var} (\varepsilon_t) \end{aligned} \quad (17)$$

- This approximated equilibrium system is now deterministic, but captures the effect of stochastics via the first two moments of the distribution of exogenous shocks

Regular perturbation second

- Then take the approximated equilibrium system

$$y_t \approx e^{\rho x_t} + \frac{1}{2} e^{\rho x_t} \text{Var}(\varepsilon_t) \quad (17)$$

- Perform Taylor approximation w.r.t. the state variable x_t about the deterministic steady state $x_t = 0$:

$$\begin{aligned} y_t &\approx 1 + \frac{1}{2} \text{Var}(\varepsilon_t) + \rho e^0 (x_t - 0) + \frac{1}{2} \rho e^0 \text{Var}(\varepsilon_t) (x_t - 0) \\ &= 1 + \frac{1}{2} \text{Var}(\varepsilon_t) + \rho \left(1 + \frac{1}{2} \text{Var}(\varepsilon_t) \right) x_t \end{aligned} \quad (18)$$

- Compare this with the true solution

$$y_t = e^{\rho x_t + \frac{1}{2} \text{Var}(\varepsilon_t)}, \quad (15)$$

whose first order approximation about $x_t = \bar{x} = 0$ is given by

$$y_t \approx e^{\frac{1}{2} \text{Var}(\varepsilon_t)} + e^{\frac{1}{2} \text{Var}(\varepsilon_t)} \rho x_t \quad (19)$$

- Stochastics should affect the mean and the slope!

Regular perturbation second

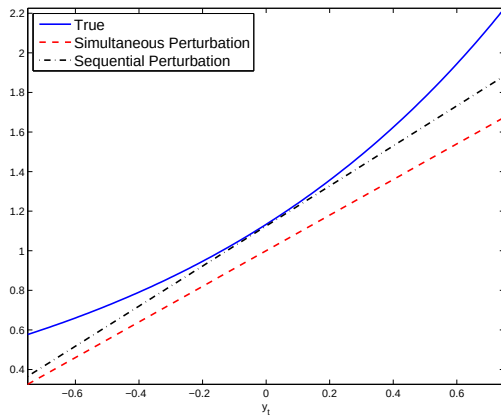


Fig. 1: $\rho = 0.9, \text{Var}(\varepsilon_t) = 0.25$

- But the simultaneous perturbation neglects the stochastics and results in both a mean shift and an error in the slope

Path dependency and the literature

- Various papers have tried to solve this issue (e.g. Coeurdacier et al. 2011; Juillard and Kamenik 2005; Lan and Meyer-Gohde 2013a,b)
- But none of these approaches has become the standard for various reasons

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Back to the literature on second-order perturbation

- Schmitt-Grohé and Uribe (2004) is the standard reference, but thanks to Ken Judd uses horrible tensor notation
- Gomme and Klein (2011) is more accessible due to Magnus and Neudecker (2007) matrix algebra
- We are going to follow Schmitt-Grohé and Uribe (2004), while abusing their notation to mimic a univariate case

The setup

- The equilibrium conditions of many economic models can be written as

$$E_t [f(y_{t+1}, y_t, x_{t+1}, x_t)] = 0, \quad (20)$$

where

- x_t is an $n_x \times 1$ vector of state variables (beginning of period stock)
- y_t is an $n_y \times 1$ vector of non-state variables (including controls)
- $f : \mathbb{R}^{2n_x+2n_y} \rightarrow \mathbb{R}^{n_x+n_y}$
- The system in (20) together with the transversality condition implicitly defines a solution of the form

$$y_t = g(x_t, \sigma) \quad (21)$$

$$x_{t+1} = h(x_t, \sigma) + \sigma \varepsilon_{t+1}, \quad (22)$$

where

- $\varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \Sigma)$
- σ is the **perturbation parameter**
- Note: (21)-(22) form a state-space system

Repetition: the perturbation parameter

- The Perturbation Parameter is introduced to scale the effective distribution of shocks
- Consider AR1-process:

$$z_t = \rho z_{t-1} + \sigma \varepsilon_t \quad (23)$$

- With $\sigma = 1$, the usual AR1-process results
- With $\sigma = 0$, the system becomes deterministic
- We follow the literature and treat the perturbation parameter as a state variable, i.e. perform simultaneously the singular and regular perturbation

The approximation point

- The approximation is conducted around the **deterministic steady state** characterized by $\sigma = 0$, $x_{t+1} = x_t = \bar{x}$ and $y_{t+1} = y_t = \bar{y}$ so that

$$f(\bar{y}, \bar{y}, \bar{x}, \bar{x}) = 0$$

- As we can express variables in deviation from their steady states, we can assume that $\bar{x} = \bar{y} = 0$

A useful definition

- For arbitrary functions $\tilde{g}, \tilde{h}$ define

$$\begin{aligned} z(x_t, \sigma; \tilde{g}, \tilde{h}) \\ = E_t \left[f \left(\tilde{g} \left(\tilde{h}(x_t, \sigma) + \sigma \varepsilon_{t+1}, \sigma \right), \tilde{g}(x_t, \sigma), \tilde{h}(x_t, \sigma) + \sigma \varepsilon_{t+1}, x_t \right) \right] \end{aligned} \quad (24)$$

- The exact solution by definition satisfies

$$z(x_t, \sigma; g, h) = 0 \quad (25)$$

The central idea

- The equilibrium conditions

$$\begin{aligned} F(x_t, \sigma) &\equiv E_t [f(g(h(x_t, \sigma) + \sigma \varepsilon_{t+1}, \sigma), g(x_t, \sigma), h(x_t, \sigma) + \sigma \varepsilon_{t+1}, x_t)] \\ &= 0 \end{aligned} \tag{26}$$

have to hold for all values of the states x_t

- Thus: **all partial derivatives with respect to the states have to be 0:**

$$F_{x^k, \sigma^j}(x_t, \sigma) = 0 \quad \forall x_t, \sigma, j, k, \tag{27}$$

where the subscripts indicate the k th derivative w.r.t x_t and the j th derivative w.r.t. σ

Guess and verify

- We proceed by using a **guess-and-verify** strategy
- The second-order Taylor approximation to the solution for a univariate case will take the form:

$$\hat{g}(x_t) = g_x x_t + g_\sigma \sigma + \frac{1}{2} \left[g_{\sigma\sigma} \sigma^2 + g_{xx} x_t^2 + 2g_{x\sigma} x_t \sigma \right] \quad (28)$$

$$\hat{h}(x_t) = h_x x_t + h_\sigma \sigma + \frac{1}{2} \left[h_{\sigma\sigma} \sigma^2 + h_{xx} x_t^2 + 2h_{x\sigma} x_t \sigma \right] \quad (29)$$

- The multivariate case requires more heavy notation, see Gomme and Klein (2011)
- This second order approximation can be characterized by making sure that the first two derivatives of (25) are 0, i.e.

$$Dz(0, 0; \hat{g}, \hat{h}) = 0$$

$$Hz(0, 0; \hat{g}, \hat{h}) = 0$$

- Will result in a recursive structure: first solve for first order terms, then for second order terms, etc.

Computing F_x

- Taking the derivative w.r.t. to x_t yields:

$$F_x(\bar{x}, 0) = f_{y_{t+1}}(\bar{x}, 0) g_x h_x + f_{y_t}(\bar{x}, 0) g_x + f_{x_{t+1}}(\bar{x}, 0) h_x + f_{x_t}(\bar{x}, 0) = 0 \quad (30)$$

- All the derivatives of f evaluated at the steady state are known
- Unknowns are the partial derivatives g_x and h_x
- Looks easy in the univariate case, but we are actually talking about $n \times n_x$ matrices where entry i, j is given by

$$[f_{y_{t+1}}(\bar{x}, 0) g_x h_x]_i^j = \sum_{\alpha=1}^{n_y} \sum_{\beta=1}^{n_x} \frac{\partial f^i}{\partial y_{t+1}} \frac{\partial g^\alpha}{\partial x^\beta} \frac{\partial h^\beta}{\partial x_j} \quad (31)$$

- We have a quadratic matrix equation system with $n \times n_x$ equations and $n \times n_x$ unknowns
- We know how to solve such a system, e.g. Blanchard and Kahn (1980) or Klein (2000)

Computing F_σ

- Taking the derivative w.r.t. to σ yields:

$$\begin{aligned} F_\sigma(\bar{x}, 0) &= E_t \left[f_{y_{t+1}}(\bar{x}, 0) [g_x \times (h_\sigma + \varepsilon_{t+1}) + g_\sigma] \right. \\ &\quad \left. + f_{y_t}(\bar{x}, 0) g_\sigma + f_{x_{t+1}}(\bar{x}, 0) (h_\sigma + \varepsilon_{t+1}) \right] \\ &= f_{y_{t+1}}(\bar{x}, 0) [g_x h_\sigma + g_\sigma] + f_{y_t}(\bar{x}, 0) g_\sigma + f_{x_{t+1}}(\bar{x}, 0) h_\sigma \end{aligned} \quad (32)$$

- Again, the derivatives of f evaluated at the steady state are known and so is g_x
- Thus, equation (32) is a linear homogenous system of equations in h_σ and g_σ
- If a unique solution exists, we must have

$$h_\sigma = 0 \text{ and } g_\sigma = 0$$

- This is the famous result that, at first order, the solution implies **certainty equivalence**

Computing F_{xx}

- Taking the derivative w.r.t. of F_x to x_t yields:

$$\begin{aligned} F_{xx}(\bar{x}, 0) &= [f_{y_{t+1}y_{t+1}}g_xh_x + f_{y_{t+1}y_t}g_x + f_{y_{t+1}x_{t+1}}h_x + f_{y_{t+1}x_t}]g_xh_x \\ &\quad + f_{y_{t+1}}(g_{xx}h_xh_x + g_xh_{xx}) \\ &\quad + [f_{y_t y_{t+1}}g_xh_x + f_{y_t y_t}g_x + f_{y_t x_{t+1}}h_x + f_{y_t x_t}]g_x \\ &\quad + f_{y_t}g_{xx} \\ &\quad + [f_{x_{t+1}y_{t+1}}g_xh_x + f_{x_{t+1}y_t}g_x + f_{x_{t+1}x_{t+1}}h_x + f_{x_{t+1}x_t}]h_x \\ &\quad + f_{x_{t+1}}h_{xx} \\ &\quad + f_{x_t y_{t+1}}g_xh_x + f_{x_t y_t}g_x + f_{x_t x_{t+1}}h_x + f_{x_t x_t} \end{aligned} \tag{33}$$

- Makes use of product and chain rule
- This is a $n \times n_x \times n_x$ system of linear equations in as many unknowns

- Next: Taking the derivative w.r.t. to x of

$$F_{\sigma}(\bar{x}, 0) = E_t \begin{bmatrix} f_{y_{t+1}}(\bar{x}, 0) [g_x h_{\sigma} + g_x \varepsilon_{t+1} + g_{\sigma}] \\ + f_{y_t}(\bar{x}, 0) g_{\sigma} + f_{x_{t+1}}(\bar{x}, 0) (h_{\sigma} + \varepsilon_{t+1}) \end{bmatrix}$$

- We have

$$\begin{aligned} F_{\sigma x}(\bar{x}, 0) = E_t \bigg\{ & f_{y_{t+1}} [g_{xx} \textcolor{red}{h}_{\sigma} + g_x h_{\sigma x} + g_{xx} \textcolor{red}{\varepsilon}_{t+1} + g_{\sigma x} h_x] \\ & + f_{y_t} g_{\sigma x} \\ & + f_{x_{t+1}} h_{\sigma x} \bigg\} = 0, \end{aligned} \tag{34}$$

where red terms are 0

- This is a linear homogenous system of $n \times n_x$ equations in the same number of elements

Computing $F_{\sigma x}$

- For

$$F_{\sigma x}(\bar{x}, 0) = f_{y_{t+1}}[g_x + h_x] h_{\sigma x} + f_{y_t} g_{\sigma x} + f_{x_{t+1}} h_{\sigma x}$$

to hold for all x_t and the solution being unique, we must have

$$h_{\sigma x} = 0 \tag{35}$$

$$g_{\sigma x} = 0 \tag{36}$$

- The volatility of the system does not affect the response to the state variables at second order

Computing $F_{\sigma\sigma}$

- Next: Taking the derivative w.r.t. to σ of

$$F_{\sigma}(\bar{x}, 0) = E_t \left[\begin{array}{l} f_{y_{t+1}}(\bar{x}, 0) [g_x h_{\sigma} + g_x \varepsilon_{t+1} + g_{\sigma}] \\ + f_{y_t}(\bar{x}, 0) g_{\sigma} + f_{x_{t+1}}(\bar{x}, 0) (h_{\sigma} + \varepsilon_{t+1}) \end{array} \right]$$

- We have

$$\begin{aligned} F_{\sigma\sigma} = & E_t \left\{ f_{y_{t+1}} \left[\underbrace{g_x h_{\sigma\sigma} + g_{x\sigma} h_{\sigma}}_{(g_x h_{\sigma})'} + \underbrace{(g_{xx} (h_{\sigma} + \varepsilon_{t+1}) + g_{x\sigma}) \varepsilon_{t+1}}_{(g_x \varepsilon_{t+1})'} + g_{\sigma\sigma} \right] \right. \\ & + f_{y_{t+1}} \cdot [g_x h_{\sigma} + g_x \varepsilon_{t+1} + g_{\sigma}] \\ & + f_{y_t} g_{\sigma\sigma} + f_{y_t} \cdot g_{\sigma} \\ & + f_{x_{t+1}} \cdot [h_{\sigma} + \varepsilon_{t+1}] \\ & \left. + f_{x_{t+1}} h_{\sigma\sigma} \right\}, \end{aligned} \tag{37}$$

where red terms are 0

Computing $F_{\sigma\sigma}$, continued

- Evaluating the cross-derivatives of f yields:

$$\begin{aligned} F_{\sigma\sigma} = E_t \bigg\{ & f_{y_{t+1}} \left[g_x h_{\sigma\sigma} + g_{xx} \varepsilon_{t+1}^2 + g_{\sigma\sigma} \right] \\ & + (f_{y_{t+1}y_{t+1}} g_x \varepsilon_{t+1} + f_{y_{t+1}x_{t+1}} \varepsilon_{t+1}) [g_x \varepsilon_{t+1}] \\ & + f_{y_t} g_{\sigma\sigma} \\ & + (f_{x_{t+1}y_{t+1}} g_x \varepsilon_{t+1} + f_{x_{t+1}x_{t+1}} \varepsilon_{t+1}) [\varepsilon_{t+1}] \\ & + f_{x_{t+1}} h_{\sigma\sigma} \bigg\}, \end{aligned} \tag{38}$$

where

$$E_t[\varepsilon_{t+1}\varepsilon_{t+1}]$$

corresponds to the covariance matrix of shocks

- This is a system of n equations in n unknowns $h_{\sigma\sigma}$ and $g_{\sigma\sigma}$

The solution

- Summarizing, we have that all odd derivatives w.r.t. to the perturbation parameter σ are zero
- Thus, the second-order solution is given by

$$\hat{g}(x_t) = g_x x_t + \frac{1}{2} \left[g_{\sigma\sigma} \sigma^2 + g_{xx} x_t^2 \right] \quad (28)$$

$$\hat{h}(x_t) = h_x x_t + \frac{1}{2} \left[h_{\sigma\sigma} \sigma^2 + h_{xx} x_t^2 \right] \quad (29)$$

- At second-order, the solution to a stochastic DSGE model of the class considered here, differs from a deterministic setup only in a constant term $0.5g_{\sigma\sigma}\sigma^2$

Stochastic volatility

- Here, we only considered homoskedastic shocks, so **stochastic volatility** of the form

$$x_t = \rho x_{t-1} + e^{\sigma_t} \nu_t, \quad \nu_t \sim \mathcal{N}(0, 1) \quad (39)$$

$$\sigma_t = (1 - \rho^\sigma) \bar{\sigma} + \rho^\sigma \sigma_{t-1} + \eta \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1) \quad (40)$$

is excluded

- Fernández-Villaverde, Guerrón-Quintana, and Rubio-Ramírez (2010) show that the previous results also hold true with stochastic volatility
- The perturbation parameter will only have coefficients different from zero in even powers
- Volatility shocks appear with coefficients different from zero only in the term where they are multiplied by the innovation to their own structural shock

⇒ Pure volatility shocks have no effect up to second order

Higher-order approximations

- Previous procedure can be easily adapted to higher orders
- Given second order terms, third order derivatives yield linear equation system that can be easily solved $\rightarrow$ Dynare does this for you at arbitrary order
- Most important term that appears at third order: $g_{x\sigma\sigma}$
- Recap:
 - First order: certainty equivalence
 - Second order: constant term in policy function
 - Third order: volatility shocks enter the policy function linearly

The Stochastic Steady State

- The deterministic steady state $\bar{x}$ is the fixed point of the deterministic model (i.e. in the total absence of shocks):

$$f(\bar{y}, \bar{y}, \bar{x}, \bar{x}, \sigma = 0) = 0$$

- However: in stochastic economy, even in the absence of current shocks, agents are aware of possibility of future shocks hitting the economy
→ will adjust their behavior to reflect this
- **stochastic steady state** $\tilde{x}, \tilde{y}$ is the fixed point of the stochastic model in the absence of future shocks

$$E_t [f(\tilde{y}, \tilde{y}, \tilde{x}, \tilde{x}, \sigma = 1, \{\varepsilon_i\}_{i=t}^{\infty} = 0)] = 0 \quad (41)$$

- Traditional first order approximation: both concepts coincide
- Not the case anymore at higher order

Stochastic steady state and ergodic mean

- Stochastic RBC model: agents have precautionary motive
→ capital stock higher than in deterministic world
- Term $h_{\sigma\sigma}$ embeds part of the precautionary behavior of agents:

$$\hat{h}(x_t) = h_x x_t + \frac{1}{2} \left[h_{\sigma\sigma} \sigma^2 + h_{xx} x_t^2 \right] \quad (29)$$

- If started at $x_t = \bar{x} = 0$: if there are no realized shocks, system will move away from deterministic steady state and wander to stochastic steady state
- Note: generally also different from **mean of the ergodic distribution**, which is the unconditional expectation of the endogenous variables
- If there are shock realizations, due to Jensen's Inequality and curvature in the policy function, those shock realizations (zero for the stochastic steady state) matter

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The problem with IRFs

- With a non-linear solution, traditional IRFs do not work anymore
- The IRFs depend on two things:
 1. the sequence of future shocks, u_t and
 2. the point in the state space at which the IRFs are started, i.e. the past history of shocks, Ω_t .
- Koop et al. (1996) suggested the concept of Generalized IRFs (GIRF) that e.g. allow considering “representative” IRFs at the ergodic mean
- The GIRF at time $t + n$ after a shock u_t is given by

$$GIRF_n(u_t, \Omega_{t-1}) = E[Y_{t+n}|u_t, \Omega_{t-1}] - E[Y_{t+n}|\Omega_{t-1}] , \quad (42)$$

(or some variation of this)

- Given a point in the state space, the future shock realizations are averaged out

How to generate GIRFs?

- Additional problem: without pruning, IRF simulations can explode but with pruning one needs initial conditions not just for economic states, but for the whole augmented state space
- There are infinitely many combinations of first to n th order terms that could have led to a particular value of the economic states
- There are two special points one can consider:
 1. The ergodic mean where all components are at their expected values. i.e. the fixed point of the stochastic pruned solution
 2. The stochastic steady state, i.e. the fixed point of the deterministic pruned solution

How to generate GIRFs?

- Andreasen et al. (2018) provide an analytical way to generate GIRFs from the pruned third order state space, starting at the ergodic mean
- Dynare uses simulation-based approach to achieve the same result
- The GIRFs can be computed from the difference between two simulated series:
 1. a simulated series of 140 periods using random shocks where at time $t = 100$ an additional 1 standard deviation shock is added to whatever shock ε_{100} was already drawn
 2. a simulated series of 140 periods using the same random shocks, but without the additional 1 standard deviation shock at time $t = 100$
- By repeating this several times and taking the average, the effect of the random shocks is averaged out ($\approx$ Monte Carlo Integration)
- For volatility shocks, a lot of repetitions may be necessary

Fernández-Villaverde, Guerrón-Quintana, Rubio-Ramírez, and Uribe (2011)

IRFs

- Fernández-Villaverde, Guerrón-Quintana, Rubio-Ramírez, and Uribe (2011) do not compute GIRFs in the classical sense
- They also condition on future shocks by setting them to 0 and starting the IRFs at the **ergodic mean in the absence of shocks/stochastic steady state**
- Denote the future realization of shocks with Ω^{fut} . The definition used is

$$\begin{aligned} IRF_n(\nu_t, \Omega_t) = & E \left[Y_{t+n} | u_t, \Omega_{t-1} = \{\dots, 0\}, \Omega_{t+1}^{fut} = \{0, \dots\} \right] \\ & - E \left[Y_{t+n} | 0, \Omega_{t-1} = \{\dots, 0\}, \Omega_{t+1}^{fut} = \{0, \dots\} \right], \end{aligned} \quad (43)$$

where the expected values can be dropped as everything is deterministic

→ no repetitions necessary

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